BILINGUAL STUDY EDITION

Introduction

引言 · 英中对照译文

编排方式：每段英文原文后紧接中文译文。正文、章节标题、图注及脚注均已收录；原文件中重复出现的页眉、页码和 DOI 页脚已省略。两幅原图按原文位置重现。

Source: introduction.pdf · 22 pages

Prepared for reading and seminar use

Introduction

引言

ENGLISH ORIGINAL

> *Undoubtedly his greatest contributions were in the field of psycho-history. Seldon found the field little more than a set of vague axioms; he left it a profound statistical science. Isaac Asimov, Foundation (1942)*

中文译文

> 毫无疑问，他最伟大的贡献是在心理史学领域。塞尔登初入这一领域时，它不过是一些模糊公理的集合；而他离开时，已将其发展为一门深刻的统计科学。——艾萨克· 阿西莫夫，《基地》（1942）

ENGLISH ORIGINAL

Danger is implicit even in the metaphor. Mining refers specifically to the extraction of valuable metallic ores from the earth, and metaphorically to any process that extracts the rare and valuable content from its surrounding context, usually at the cost of extreme expenditures of labor and at danger to the environment. Data mining, a similarly intensive operation with implicit hidden costs, also begins with counting, but includes statistical transformations to test for relationships such as correlation and significance. Text mining, a process with all the dangers of the former two categories, mines for value with text as the data.

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危险甚至隐含在这一隐喻之中。“采矿”具体指从地下开采有价值的金属矿石；在比喻意义上，它泛指从周围语境中提取稀有而有价值的内容，通常要付出极高的劳动成本，并对环境造成危害。数据挖掘同样是一种高强度、带有隐性成本的操作：它也从计数开始，但还包括统计转换，以检验相关性、显著性等关系。文本挖掘兼具上述两类活动的所有风险，它把文本作为数据，从中发掘价值。

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It begins with computational transformations that break up and classify digitalized strings of archival text into units representing constituent words and phrases, and then applies statistical manipulations that enable us to study the kinds of meaningful signals and relationships that trained analysts from the humanities and social sciences would like to detect in the course of reading. With the help of computers, the insights of reading can be carried out over a larger scale. Text mining is powerful if done right. When rendered as visualizations, the output of text mining offers, at least potentially, an outsized return on investment, distilling shelf miles of text into a valuable, pithy representation of what all those words meant. A single visualization might reduce the story of how a single institution’s politics have changed over a hundred years.

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它从计算转换开始，将档案文本的数字化字符串分解和分类为代表组成词和短语的单位，然后运用统计操纵，使我们能够研究从人文和社会科学中培训分析人员在阅读过程中想要发现的各种有意义的信号和关系。在计算机的帮助下，阅读的洞察力可以在更大的范围内进行。文本挖掘如果做得对的话是强大的。 文本挖掘的输出作为可视化，至少有可能带来投资的超额回报，将书架里程的文本提炼成一个有价值的、细微的表述所有这些词的含义。 一个单一的视觉可能减少一个机构的政治在一百年中是如何变化的故事。

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Scholars from the humanities have used text mining to give us a mirror of what stories people have told each other about how Covid is transmitted, and how American novelists present white characters and black characters differently over time. We see an example of this method in Figure 0.1, where word embeddings have been used to show how members of the US Congress talked about the term “slavery” from 1870 to 2010. The x-axis is a timeline from 1870 to 2020 (the full length of the dataset), divided into five-year periods. The y-axis gives an abstract measure of “similarity” between the keyword “slavery” and words that appeared in the same sentence, paragraph, or debate. Words higher on the y-axis appeared alongside “slavery” with greater regularity. The diagram was designed to show the shifting context in which the word “slavery” appeared from one generation to the next.

中文译文

人文学科的学者们用文本挖掘来给我们一个镜子，描绘人们互相讲述的新冠病毒如何传播的故事，以及美国小说家如何随着时间推移而不同地呈现白人人物和黑人人物。我们从图0.1中看到这种方法的一个例子，其中使用了词嵌入来显示美国国会成员在1870年至2010年期间如何谈论“奴隶”一词。 x轴是1870年至2020年(数据集全长)的时长，分为5年. y轴对关键词“奴隶制”和同一句、段落或辩论中出现的词语之间的“相似性”作了抽象的衡量。 在Y轴上高调的字句与“奴隶”并列出现，频率更高。 该图旨在显示＂奴隶＂一词从一代到下一代出现的不断变化的背景。

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The big takeaway of Figure 0.1 is that for most of the period since the Civil War, Congress has not talked about race when it has spoken about slavery. Members of Congress used the term “slavery” as a metaphor for a host of other ills they wished to denounce. They classified Mormon polygamy, Naziism, and Bolshevism forms of mental or moral enslavement. But much less frequently did members of Congress acknowledge that slavery as an institution had wrecked the lives of generations of African-Americans, brought to the United States against their will. Nor did members of Congress talk about the legacy of slavery in modern America. Instead, they associated slavery with developments abroad, especially the “economic slavery” that they associated with communist regimes. Congress’s reluctance to talk about the facts of American history in the chief organ of American democracy persisted for over a century.

中文译文

图0.1的主要取材是，自内战以来的大部分时间里，国会在谈论奴隶制时没有谈到种族。国会议员们用“奴隶制”一词来比喻他们希望谴责的许多其他弊端。 他们将摩门教一夫多妻，纳粹主义，和布尔什维克式的精神或道德奴役形式归类为一类。但国会成员很少承认，奴隶制作为一个机构破坏了几代非裔美国人的生活，违背他们的意愿带到美国。国会成员也没有谈论现代美国奴隶制的遗留问题。相反，它们把奴隶制与国外的发展联系在一起，尤其是与共产主义政权有关的“经济奴隶制”。 国会不愿在美国民主的主要机构谈论美国历史的事实，这种情况持续了一个多世纪。

ENGLISH ORIGINAL

Only in the 1990s did “racism” and “lynching” become collocates of “slavery” for the first time. A diagram like Figure 0.1 is designed to dramatize shifts over time, shifts that may be so subtle that they surprise many an American historian. 2 It is not the algorithm alone that renders such an insight obvious. The algorithm has to be deployed by a coder who understands the significance of change over time to patterns of words that appear with other words. Unlocking the mysteries of the algorithm, the category of time reveals how we have lied to each other about our institutions and identity. There are also many data-driven analyses of text that purport to offer substantive insight - and fail. Such failures of text mining occur when algorithmic distillations of text are misapplied, with the result of analyses that are empty, biased, or simply false. Consider the fate of the wordcloud in recent years.

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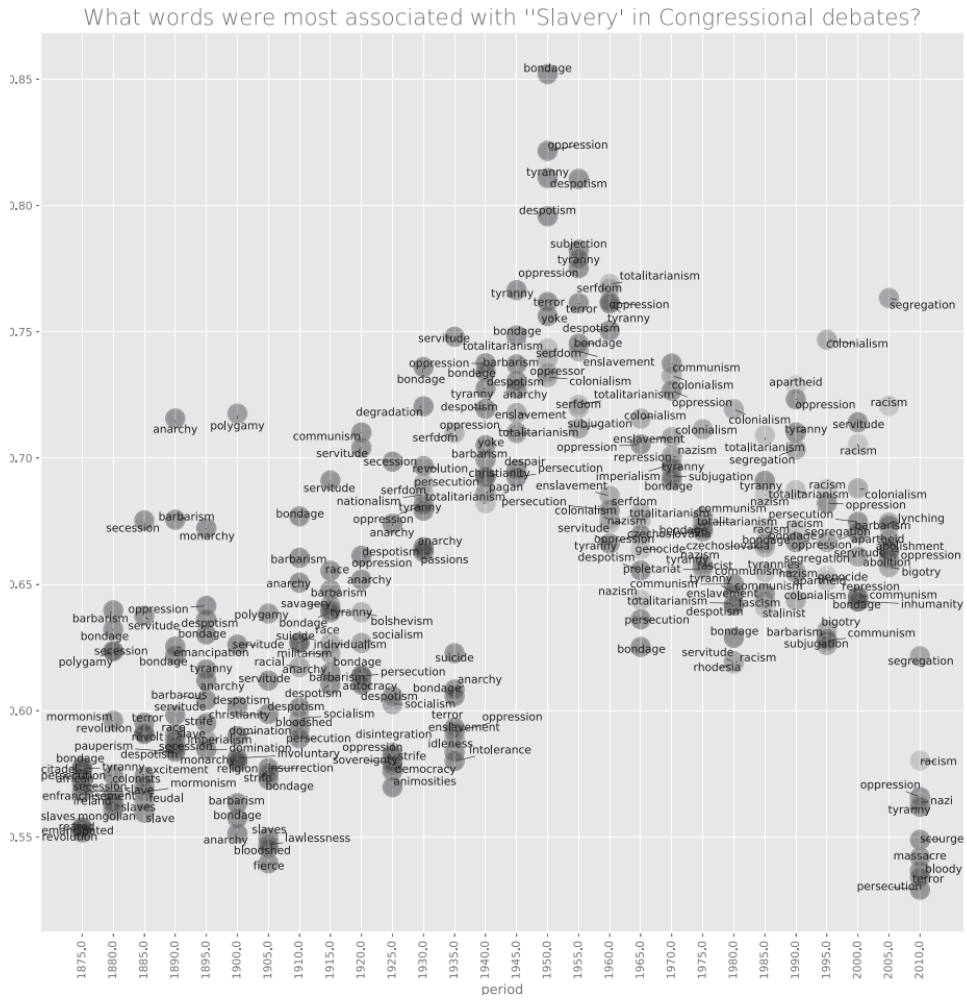
只是在1990年代，" 种族主义 " 和 " 私刑 " 才第一次成为 " 奴隶 " 的集合体。 像图0.1这样的图表 旨在戏剧化 变化随着时间，变化可能如此微妙，2个 使这种洞察力显而易见的不仅仅是算法。 该算法必须由一个能理解随时间推移而变化对用其他词出现的词的规律的意义的编码器来部署。在解开算法的奥秘后，时间的类别揭示了我们如何对我们的机构和身份相互欺骗。 此外，还有许多数据驱动的文本分析，这些分析旨在提供实质性的洞察力 -- -- 而失败。 这种文本挖掘的失败发生在文本的算法衍射被错误应用时，其分析结果是空的，偏颇的，或者纯粹的虚假的。考虑近年来"云"字的命运.

ENGLISH ORIGINAL

The wordcloud (sometimes known as a “wordle” before the arrival of a popular game of the same name) is a visualization where the size of words corresponds to their frequency. In the wordcloud visualization in Figure 0.2, the larger words are at the center, but the words’ spatial arrangement is random, a stylistic choice that encourages the viewer’s eyes to wander the cloud, free associating along the way.

中文译文

词云(有时在同名的流行游戏到来前称为"词")是一种可视化，词的大小与其频率相对应。在图0.2中的词云可视化中，较大的单词位于中心，但单词的空间安排是随机的，是一种鼓励观众眼游云，沿途自由联动的风格选择.



ENGLISH ORIGINAL

Figure 0.1 Words co-located with the keyword “slavery” in the debates of the US Congress. X-axis indicates time; y-axis indicates an abstract measurement of “similarity” measuring how consistently the words are invoked together. Detected using Gensim word embedding software.

中文译文

图 0.1 美国国会辩论中与关键词“奴隶制”共同出现的词语。横轴表示时间；纵轴表示“相似度”的抽象测量，即这些词语共同被使用的一致程度。使用 Gensim 词嵌入软件检测。

ENGLISH ORIGINAL

After first appearing as a navigation tool on the photo-sharing website Flickr in 2004, wordclouds became a preferred visualization used by newspapers for reducing political speeches into a pop graphic. From 2004 to 2017, the wordcloud was everywhere. Bloggers and websites used wordclouds as a navigation index. During the Obama presidency, newspapers such as The Guardian regularly ran visualizations distilling Barack Obama’s State of the Union speeches into a wordcloud. But the wordcloud abruptly disappeared from newspapers and blogs alike when political events tested the limits of the tool to reveal the truth. As late as the Obama presidency, the wordcloud served the public as an easy tool for rendering complex text into a simple snapshot of a historical moment.

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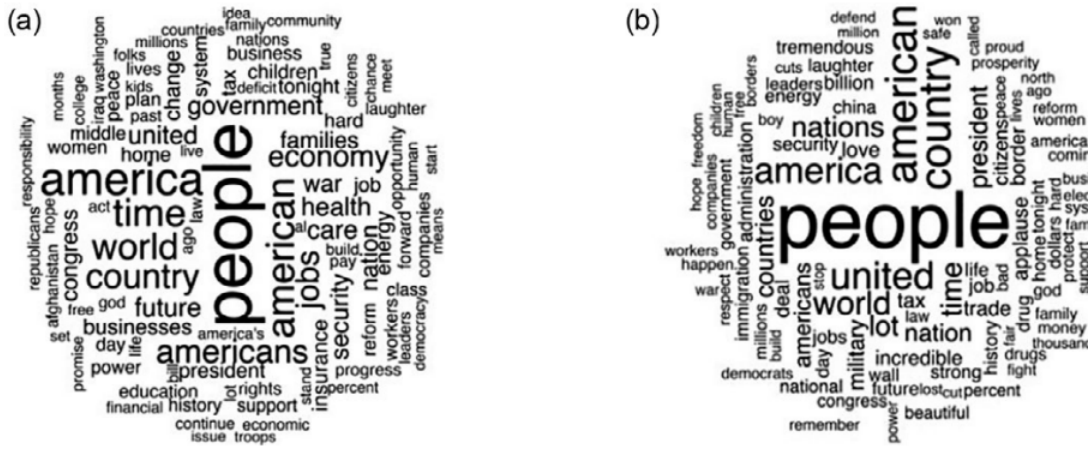
在2004年首次作为导航工具出现在照片分享网站Flickr上后，wordclouds成为报纸用来将政治演讲简化为流行图形的首选可视化。从2004年到2017年，词云无处不在。博客和网站使用文词云作为导航索引。在奥巴马任总统期间，《卫报》等报纸经常将巴拉克· 奥巴马的国情演讲提炼成文词云。 但当政治事件考验了揭示真相工具的限度时，早在奥巴马担任主席期间，文词云就为公众服务，成为将复杂文本变成一个历史时刻的简单快照的简单工具。

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In Obama's speeches, keywords such as "freedom,""people," and "journey" loomed large in size, reinforcing Obama's sense of America as a nation at a turning-point in history. 3 Just as importantly, the wordcloud promised to distill into a single visualizable truth the reality of truth in an era of information overload. Soon after Donald Trump's election, however, the fad for wordclouds began to dissipate. Under a new president, language was being used in a fundamentally new way, but wordclouds failed to do justice to the difference. Indeed, had journalists in 2017 the power to compare all of Trump's speeches from his presidency with Obama's, they would have discovered a

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在奥巴马的演讲中，诸如“自由”、“人民”和“旅程”等关键词在规模上浮现，强化了奥巴马在历史转折点对美国的看法。 3 同样重要的是，文词云承诺在一个信息超载的时代，将真相的真相提炼成单一可视的真理。 然而，唐纳德· 特朗普当选后不久，热衷于文词云的潮流开始消散。在新总统统治下，语言被以一种全新的方式使用，但文词云未能公正对待分歧。 事实上，如果记者在2017年有权将特朗普总统的所有演讲与奥巴马的演讲进行比较，他们就会发现



ENGLISH ORIGINAL

Figure 0.2 Data-driven visualizations that obscure rather than illuminate meaningful differences: Wordclouds for the official speeches of a) Barack Obama and b) Donald Trump during their presidencies. Source: "Presidential Speeches" dataset, https://millercenter.org/the-presidency/presidential-speeches. Visualized by author.

中文译文

图 0.2 非但未能阐明、反而遮蔽了重要差异的数据驱动可视化：a）巴拉克· 奥巴马与 b）唐纳德· 特朗普在总统任期内官方演讲的词云。来源：“总统演讲”数据集，https://millercenter.org/the-presidency/presidential-speeches。由作者制作可视化。

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mysterious fact: the two men used very many of the same words at the same rate. Both proclaimed their allegiance to the "American""people" and their "country;" both referenced the "United""States." Obama talked about the "economy" and "jobs," while Trump talked about "trade" and "jobs." They both referenced "citizens,""families," and "children." Both touted their commitment to "security," to America's role in the "world," and to the importance of the "time" of the present. The more Trump spoke, the clearer the weakness of the wordcloud became. The word cloud, a tool for showing the most frequent words, did not in its automatic form look for phrases. It could not tell the difference between Obama's "right here in America"- where he intended to create jobs - and Trump's "make America great again," a plea that barely veiled a longing for a time of white supremacy.

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神秘的事实:两人以同样的速度使用了很多相同的词。两人都宣布效忠“美国”“人民”和“国家”;他们都提到“联合”国家。 奥巴马谈到了“经济”和“工作”，而特朗普则谈到“贸易”和“工作”。 他们都提到“公民”、“家庭”和“儿童”。 两者都强调对 " 安全 " 的承诺、对美国在 " 世界 " 中的作用的承诺，以及对当前 " 时代 " 的重要性的承诺。 特朗普的发言越多，词云的弱点就越明显。云字，是显示最常见词的一种工具，没有以自动的形式寻找短语。在奥巴马的“美国这里”和特朗普的“让美国再次伟大”之间，

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In the early years of the Trump presidency, journalists and politicians kept running Trump's speeches and tweets through the visualization, hoping for different results. When that approach failed, one democratic representative in Congress tried a new strategy to capture Trump's abrasive style, creating his own wordcloud generated from a list of the individuals and organizations that Trump had attacked over Twitter. 4 But mainly the wordclouds of 2017-21 continued to back fire. One published wordcloud distilled the keyword "Klansmen" as a word related to Biden, but not related to Trump. In the accompanying article, a journalist reassured his readers that future President Joe Biden had condemned the seeming condoning of the KKK by Donald Trump, who himself never mentioned the Klan by name. 5 By then, it must have become clear: Trump had broken the wordcloud.

中文译文

在特朗普总统任期的最初几年，记者和政界人士通过可视化不断主持特朗普的演讲和微博，希望得到不同的结果。当这一方法失败时，国会的一名民主代表尝试了新的策略来抓住特朗普的惨淡风格，从特朗普在推特上攻击的个人和组织名单中创造了自己的文词云. 4,但主要为2017-21的词云继续回火。一个已出版的词云将关键词"Klansmen"蒸馏成与拜登相关的词，但与特朗普无关。在随笔文章中，一名记者向他的读者保证，未来的总统乔· 拜登曾谴责唐纳德· 特朗普似乎纵容KK,他本人从未提及过三K党的名字。页:1 到那时，它肯定已经变得很清楚:特朗普已经打破了文词云.

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By aping his opponents' populist language for nationalist and racial purposes, Trump had produced a close parody of Obama's invocations of patriotism. The result was that Trump's speech was virtually indistinguishable from Obama's - at least in wordcloud form. The wordcloud betrayed how Trump's broadly inclusive language appealed to many American's sense of patriotism, indeed, wooing some voters who had cleaved to Obama, even while carefully encoded dog-whistles summoned hate groups to his side. For anyone paying attention, the wordcloud might have been trying to tell us something about the political malleability of populist language. For the purpose of explaining to newspaper readers the differences between two presidents, however, the wordcloud was kaput. Understanding the priorities, values, and falsehoods of president is a matter of no small priority. Reasonable people may reasonably disagree on political values.

中文译文

特朗普通过将对手的民粹主义语言用于民族主义和种族目的，产生了奥巴马引用爱国主义的近乎讽刺的言论。结果是特朗普的演讲与奥巴马的演讲几乎是不可区分的 — — 至少是文词云的形式。文词云背叛了特朗普广泛包容的语言如何吸引许多美国人的爱国情绪，事实上，在精心编码狗话的同时，却吸引了一些向奥巴马倾诉的选民，将仇恨团体召唤到他身边。 对于任何注意的人来说，这个词云可能一直在试图告诉我们一些关于民粹主义语言政治可塑性的问题。然而，为了向报纸读者解释两位主席之间的分歧，"云"一词已经失效。 了解总统的优先事项、价值观和谬误，是一个非常重要的优先事项。 理性的人可能在政治价值观上有合理的分歧.

ENGLISH ORIGINAL

But newspapers that serve reasonable people try to provide everyone with the same basic facts about the present moment and recent past. The wordcloud vanished because it proved incapable, at least in its automatic form, of generating straightforward facts about Trump that were capable of picking up on the president's many obvious differences from his predecessor. A world trained on the manipulation of data but ignorant of, or insensitive to, the data's original context will be dangerously cut off from reality. Text mining can aid us by helping humans become more accurate at sifting data at scale. Without the insights of the humanities, text mining falls readily into error: it is a monkey reaching for the moon, guided only by the moon's reflection in a pond, where he will never grasp it. That we should be interested in accuracy is obvious.

中文译文

但为理性人士服务的报纸试图为每个人提供同样的基本事实，说明目前和最近的历史。这个词消失了，因为事实证明，至少是自动的，它无法对特朗普产生直截了当的事实，而特朗普能够从总统的前任那里得知许多明显的分歧。 一个接受过数据操纵培训但不了解或不了解数据原始背景的世界将危险地与现实隔绝。 文本挖掘可以帮助我们帮助人类在测量数据的规模上变得更加精确。 如果没有人文学科的洞察力，文本挖掘就容易出错:它只是一只猴向月球伸展，只以月球在池塘中的反射为指南，他永远无法抓住它。 我们显然应该对准确性感兴趣。

ENGLISH ORIGINAL

Universities have long aspired to be the grounds where children reared in the native bias of their own locale learn to form an acquaintance with a broader reality. Companies that depend on the parsing of language with code should likewise be interested in the forms of hybrid thinking - humanities and mathematics, social science and informatics. Text mining can be used to paint a portrait of partisan activity in legislative debates, newspapers, and social media sites. It can also support the larger method of using data to locate a discontinuity in time and of learning as much as possible about the distinctiveness of the event relative to other similar disruptions, as well as the individuals, causes, and contexts of disruption. Once we've turned data to the purpose of understanding how two moments in time differ, we've taken a great step towards reality. We have fit data to the purpose of asking a real historical question.

中文译文

长期以来，大学一直希望成为儿童在自己地方的本地偏见中成长起来，学会认识更广泛的现实。 依赖语言与代码解析的公司同样应当对混合思维形式感兴趣——人文和数学、社会科学和信息学。 在立法辩论、报纸和社交媒体网站，它还可以支持采用更大的方法，利用数据及时确定不连续性，并尽可能了解事件相对于其他类似干扰的独特性，以及干扰的个人、原因和背景。 一旦我们把数据转换为理解时间的两瞬间如何不同的目的，我们就朝着现实迈出了一大步。 我们掌握的数据符合一个真正的历史问题的目的。

ENGLISH ORIGINAL

We've provided an instrument that is practically useful for assigning blame and pursuing reform and, even better, for starting to conceptualize the changes that determined the shape of the present. Why were solutions of this kind invisible to the staff of American newspapers during the Trump years, when the limitations of the wordcloud began to show? They were invisible because text mining for historical analysis was still in its infancy during the Trump years. The place where the new tools would appear was still a no man's land between two enemy encampments.

中文译文

我们提供了一种工具，它实际上可以用来指责和推行改革，甚至可以用来开始构思决定目前状况的变化。 为什么在特朗普年代，当文词云的局限性开始显现时，美国报纸的工作人员看不到这种解决方案? 他们被忽略了，因为用于历史分析的文本挖掘在特朗普年代仍处于初始阶段。新工具出现的地方仍然是两个敌方营地之间的无人区。

The No Man's Land Where Better Tools Come From

孕育更好工具的无人区

ENGLISH ORIGINAL

In the years of the Trump Administration, a vast canyon extended between the experts in quantitative knowledge and that collection of scholars in the liberal arts who study documents for tiny signals of cultural change over time. Let us call these individuals"The Temporal Cartographers'Society." It is they who would face no difficulty parsing the differences between Obama's speeches and Trump's. The Society's ranks include professional historians, literary scholars, and historical sociologists, as well as some of the most talented members of the world of journalism and law. Mainly, they trained in liberal arts subjects, where, if quantitative methods are used to analyze text, they have been introduced (or reintroduced) only over the last two decades. Those fields have long traditions of struggle and debate devoted to the difficulty of accurately summarizing a thousand letters.

中文译文

在特朗普政府执政的几年中，一个庞大的峡谷在数量知识专家与人文学科学者的收集之间延伸开来，这些学者研究了一段时间以来文化变化的微小信号的文件。让我们将这些人称为“时空制图师协会”。 他们不会面临任何困难，该学会的排名包括专业历史学家，文学学者，历史社会学家，以及新闻和法律界一些最有才华的成员。他们主要接受人文学科科目的培训，如果用定量方法分析文字，则仅在过去20年中才引入(或重新引入). 这些领域有着长期的斗争和辩论传统，致力于准确总结一千封信的难度。

ENGLISH ORIGINAL

Historians' standard of truth, for instance, requires them to carefully reflect on the biases and silences of the documents with which they work, to worry over the circumlocutions and hidden suggestions of text. They also worry over hidden dimensions of change to the historical context in which words are invoked, a problem not easily reduced to the words on the page. As the scholar William Sewell once explained, not only words and ideas, but also new identities structure society over time. The result is the laws of change may work differently in one generation than another. 6 For this reason, historians study the imperfections, distortions, and omissions that riddle records from the past - including circumlocutions like Trump's phrase "make America great again," a phrase that must be unpacked beyond its surface invocation of prosperity for the implied longing for an era of white supremacy and male dominance to surface.

中文译文

例如，历史学家的真理标准要求他们仔细思考他们所工作的文件的偏颇和沉默，担心文本的环绕和隐蔽建议。 他们也担心在引用词语的历史语境方面变化的隐性，这个问题不易归结为页面上的词语。正如学者威廉· 西韦尔(William Sewell)曾经解释的那样，不仅言语和思想，而且随着时间推移，新的身份结构社会。其结果是，变化法则在一代人中可能与另一代人不同。 6 为此原因，历史学家研究了谜题是过去记录的不完美、扭曲和疏漏，包括像特朗普的“让美国再次伟大”这样的环绕语句，这个短语必须摆脱表面对繁荣的援引，而去追求一个白人至上和男性统治的时代。

ENGLISH ORIGINAL

The Temporal Cartographer's Society, in other words, has a lot to offer any study that depends on absolute honesty about what we know, what we don't know, and where data has been distorted by the bias of the past. Traditionally, members of the Temporal Cartographers' Society have lacked tools for generalizing about billions or trillions of words, how those words cluster and change. They have adopted admirably to other forms of work with data - to birth and death rates, or what quantities of nutrition it was possible to buy on a working man's salary, or counting pollen samples to chart changing temperatures. 7 Even with data, however, this group tends to be hesitant about over-grand generalizations. They rarely look for laws of historical truth. When they turn to numbers, they examine moments of change with enormous precision.

中文译文

换句话说，《时空算术学会》 有许多研究可以提供，取决于我们所了解的，我们所不知道的，以及数据被过去的偏见扭曲的地方。 传统上，Temperal Cartographers学会的成员缺乏工具来概括大约数十亿或数万亿个字，这些字是如何组合和变化的。 他们对其它有数据的工作形式，如出生率和死亡率，或者用工人的工资买多少营养，或者用花粉样本来计算温度变化，都值得赞赏。 7 然而，即使有数据，这一组人往往对过于笼统地概括持犹豫态度。 他们很少寻找历史真理的法律。 当它们转向数字时，它们非常精确地审查变化的时刻。

ENGLISH ORIGINAL

Generalizations at broad scale are the domain of another group, namely the engineers, mathematicians, statisticians, and computer scientists whose conceptual domain we might call the "world without history," because thinking about cultural and political revolution represents so little of the work done in that domain. The world without history is the imaginary territory that is home to students of statistics and machine learning. Their labors support the design of computers and the creation of a thousand tools for animating design, medicine, and neuroscience. And some of its scions apply math on the stock market, where they sometimes leverage historical data while promising to predict the future. The world without history has its own blindspots. In my experience, few computer scientists are exposed to historicist reasoning anywhere in their training.

中文译文

广义的概括是另一个集团的范畴，即工程师、数学家、统计学家和计算机科学家，他们的概念领域我们可以称为“没有历史的世界”,因为对文化和政治革命的思考很少代表在这一领域所做的工作。 没有历史的世界是统计和机器学习学生的想象中领土。 他们的劳动支持计算机的设计，并为动画设计，医学，神经科学创造了一千种工具。而它的一些学说在股市上应用了数学，它们有时利用历史数据，同时有望预测未来。没有历史的世界有自己的盲点。 在我的经验中，很少有计算机科学家在训练中接触到历史主义推理.

ENGLISH ORIGINAL

They often lack the broad literacy in the humanities one needs to understand the potential and the limitations of these conceptual models. And this ignorance of computationalists has made trade - let alone conversation - between the world without history and the Temporal Cartographers' Society virtually nonexistent for many years. The Temporal Cartographers' Society entertains a certain skepticism towards the world without history. Issues of race, gender, the silences of the archive, and the costs of labor highlight the exploitative implications latent in the phrase "text mining," which refers to the process of computationally extracting meaning from digitalized documents.

中文译文

他们往往缺乏了解这些概念模式的潜力和局限性所需的人文知识。 而这种对计算家的无知使得没有历史的世界和时空制图家之间的贸易，更不用说对话了。 社会实际上已不存在多年。 漫画家协会对没有历史的世界持某种怀疑态度。 种族、性别、档案的沉默和劳动成本等问题突出了“文本挖掘”一词潜在的剥削性影响，它指的是从数字化文件中计算出意义的过程。

ENGLISH ORIGINAL

As Kate Crawford has pointed out, the metaphor of "mining" underscores the promise that computational labor will yield precious resources and therefore merits corporate investments. 8 But as historians know, mining industries have also been classical sites of the exploitation of indigenous, ethnic, and other working-class labor, entailing the making invisible of laboring bodies often broken, suffocated, and poisoned by working conditions. Mining industries have also been sites of lasting environmental pollution, as well as the origin point of the fossil economies, dependence on which has produced the carbon emissions of recent centuries, resulting in the climate emergency. To insist on referring to the work of computational history with texts as "text mining" is to testify to the fact that computational enterprises are implicated in potentially exploitative relationships with labor and the environment.

中文译文

正如凯特· 克劳福德所指出，"采矿"的比喻凸显出计算劳动力将产生宝贵资源从而值得公司投资的希望. 8 但正如历史学家所知，采矿业也一直是剥削土著、族裔和其他工人阶级劳工的古典场所，导致劳动身体经常被打破、窒息和被工作条件毒害。 采矿业也是长期环境污染的地点，也是化石经济的发源地，依赖化石经济产生了近几个世纪的碳排放，导致气候紧急情况。 坚持将文本中的计算历史工作称为“文本挖掘”,就是证明计算企业与劳工和环境之间可能存在剥削关系。

ENGLISH ORIGINAL

There is a longer story behind the distance between the two worlds as well. In American history departments, many faculty regard with distaste previous encounters with the methods of the world without history. The socalled quantitative turn of the 1960s and 1970s is remembered today as a moment when millions of dollars supported vast research collaborations in history and adjacent social science fields. The Philadelphia Project, a study of social mobility and the American Dream, is remembered chie flya sa n expensive disaster whose authors failed to publish meaningful work.

中文译文

在两个世界之间的距离后面还有一个较长的故事。 在美国历史系中，许多院系都认为以前对没有历史的世界方法的遭遇是厌恶的. 1960年代和1970年代的所谓定量转折今天被人们记住，当时有数百万美元支持了历史和相邻社会科学领域的广泛研究合作。 费城计划是对社会流动性和美国梦的研究，被人们记得是昂贵的灾难，

ENGLISH ORIGINAL

More notorious still is Robert Fogel and Stanley Engerman's Time on the Cross, which used a data base of life-expectancy to argue that enslaved Americans were treated better than their working-class counterparts in the north. 9 The numbers in Fogel and Engerman's project included 0.7 as the number of times that the average enslaved person was whipped per year at a single plantation in Louisiana between 1840 and 1842. 10 The crudely racist reductionism associated with these projects - which offered a decimal in the spot that could be occupied by tomes on the separation of families of enslaved people, the experience of the body in pain, and intergenerational trauma - still circulates in history departments today as evidence of what can go wrong when the desire for quantitative accuracy overrides humanists' appreciation of the human condition.

中文译文

更臭名昭著的是罗伯特· 福格尔(Robert Fogel)和斯坦利· 恩格曼(Stanley Engerman)的"十字架上的时光"(Time on the Cross),后者利用生命期望数据库来论证被奴役的美国人比北方的工人阶级的对手受到更好的待遇。第9条 福格尔和恩格曼项目中的数字包括0.7作为1840年至1842年间平均被奴役者在路易斯安那州单一种植园每年被鞭打的次数. 10 与这些项目有关的粗鲁的种族主义减少主义——在被奴役者家庭分离、痛苦中的身体经历和代际创伤等方面可能占到小数点——今天仍然在历史部门流传，以证明当对数量准确性的渴望压倒了人类主义者对人类状况的理解时，会发生什么错误。

ENGLISH ORIGINAL

Even as nearby fields such as political science and sociology have embraced big data, with scholars refining the methods that illuminate the major theories of the field, the refinement of similar tools for dealing with big data in history has only just begun. As recently as the Trump years, historians' warnings about the dangers of quantitative tools and information scientists' patterns of selective study meant that a dangerous and underexplored territory sprawled between the world without history and the domain of the Temporal Cartographers' Society. Few travelers crossed from one domain to the next. Much rarer was a compelling paper embraced by both worlds as the signal of a new field ripe for investigation. Today, promising things are afoot in this no man's land.

中文译文

即便近代政治学和社会学等领域都接受了大数据，学者们重新定义了阐明该领域主要理论的方法，但处理历史上大数据的类似工具的重新精细化也刚刚开始。最近，在特朗普年代，历史学家对定量工具和信息科学家选择性研究模式的危险性的警告意味着，一个危险和未充分探索的领土在没有历史的世界和Temperal制图家协会的领地之间蔓延。 很少有旅行者从一个域穿越到另一个域。更罕见的是，这两个世界都接受了一份令人信服的文件，作为调查时机成熟的新领域的信号。 今天，在这片无人居住的土地上，充满希望的东西已经浮现。

ENGLISH ORIGINAL

Modern "digital historians" who use text mining in their work, for instance Luke Blaxill, readily credit the linguistic turn as the inspiration for their own careful use of wordcount and statistics to discern moments of change in political speeches. The space of "digital history" that applies computational tools to historical problems only now has its first journals - Current Research in Digital History was founded in 2018 and the Journal of Digital History was founded in 2021. Luke Blaxill's 2020 book, The War of Words, is the first historical monograph to use text mining as its major method. The truth is that scholars in the liberal arts are continuing to profit from and learn from debates about quantitative reasoning in the past, and they have many insights to offer those who work with data.

中文译文

现代“数字历史学家”在他们的工作中使用文本挖掘，例如Luke Blaxill,他们很容易将语言转向作为他们自己仔细使用文字统计和统计的灵感来识别政治演讲中的变化时刻。 将计算工具应用于历史问题的"数字历史"空间现在只有它的第一份期刊——"数字历史的当前研究"成立于2018年，"数字历史学报"成立于2021年. Luke Blaxill的2020年著作"文字之战"是第一部以文本挖掘为主要方法的历史专著。事实是，文艺界的学者们继续从过去关于定量推理的争论中获利并从中学习，他们有许多见解可以向那些从事数据工作的人提供.

ENGLISH ORIGINAL

African-American and other postcolonial scholars - especially Jessica Marie Johnson and Roopika Risam - have taken a leading role in describing the possible abuses of quantified perspectives on the experiences of enslaved or otherwise silenced and exploited communities in the past. 13 Distant reading and the promise of digitalized data have attracted the attention of historians since at least Lincoln Mullen and Keller Funk's "Spine of the American Law" (2016), where legal historians adopted algorithms to detect important turning-points in the history of American case law.

中文译文

非裔美国人和其他后殖民学者，特别是杰西卡· 玛丽· 约翰逊和罗皮卡· 里萨姆，在描述过去对被奴役或以其他方式压制和被剥削社区的经历可能滥用量化观点时发挥了主导作用。 13 异形读数和数字化数据的承诺引起了历史学家的注意，因为至少林肯· 穆伦(Lincoln Mullen)和凯勒· 丰克(Keller Funk)的"美国法的松"(2016年),法律史学家在其中采用了算法来探测美国判例法史上的重要转折点.

ENGLISH ORIGINAL

Since that point, the range of algorithmic tools for textual analysis available to digital historians has expanded enormously, and a thriving community of digital historians have begun to investigate the best practices of text mining. 14 Researchers in History have begun to apply artificial intelligence (AI) tools to the history field, and my publications have explored a range of different algorithmic approaches. Many of my own technical articles demonstrate positive proof of the effectiveness of natural language processing (NLP) and statistical measures for discovering hitherto unknown historical events in the history of capitalism. 15 The research is truly "hybrid" in the sense of representing a new undertaking that fits neither into traditional categories of history nor of information science.

中文译文

从那时起，数字历史学家可利用的文字分析算法工具范围大大扩展，一个兴旺的数字历史学家群体开始调查文本挖掘的最佳做法. 14 历史研究者开始将人工智能(AI)工具应用于历史领域，我的出版物探索了一系列不同的算法方法。我本人的许多技术文章都证明了自然语言处理(NLP)和统计措施对于发现资本主义历史上迄今未知的历史事件的有效性。 15 这一研究真正是 " 杂乱无章 " 的，它代表着既不符合传统历史类别也不符合信息科学的新事业。

ENGLISH ORIGINAL

This research requires scholars to formulate new methods for validating whether and how each AI tool can support the high standard of reasoning required by the history profession. 16 This body of work, as a whole, points the way towards new standards of believability as well as new opportunities for building and measuring historically significant moments whose traces are embodied in textual data. The expertise of the new hybrid field is sometimes called the "digital humanities" or "the digital social sciences"; and one corner of it - trained on pattern-recognition in images, movement, maps, video, audio, and text alike - is sometimes described as "cultural analytics."17 The expertise in text mining in particular commands a certain corner of cultural analytics, and the problem of historical truth adds another still more specific dimension of concern. But history is not of concern only to historians.

中文译文

这种研究要求学者们制定新的方法来验证每个AI工具是否能够以及如何支持历史专业所要求的高标准推理. 16 整个这一组工作指明了通向新的可信标准的道路，以及构建和衡量历史上重要时刻的新机会，其痕迹体现在文字数据中。 新的混合领域的专门知识有时被称为"数字人文"或"数字社会科学";其中一个角落——在图像、运动、地图、视频、音频和文本中都受过模式识别培训——有时被称为"文化分析"。 文本挖掘方面的专门知识尤其占据了某种文化分析的一角，历史真相问题又增加了另一个更具体的关切问题。 但历史不只是历史学家所关心的.

ENGLISH ORIGINAL

It is also a method embraced across the humanities and social sciences, relevant to law schools and schools of journalism, and useful in practically any endeavor where the investigation of human culture and institutions is at stake. I prefer to refer to the activities covered by this book as “text mining for historical analysis.” “Text mining” is a province of research in AI, and there are researchers in computer science and informatics departments dedicated to the question of what makes text mining accurate and robust. Its practices borrow richly from the interdisciplinary cultural analytics community as well as practitioners of text mining who work across the social sciences, especially in political science and sociology.

中文译文

它也是贯穿人文和社会科学的一种方法，它与法学院和新闻学院有关，在涉及人类文化和机构调查的实际上任何努力中都是有用的。 我更愿意将本书所涵盖的活动称为“历史分析的文本挖掘”。 “文字开采”是大赦国际的一个研究领域，计算机科学和信息学部门中有研究人员专门研究如何使文字开采准确和有力。 其做法大量借鉴了跨学科文化分析界以及从事跨社会科学，特别是政治科学和社会学工作的文本开采业者。

ENGLISH ORIGINAL

“Historical analysis” marks out diachronic text mining, trained on the modelling of temporal experience in all its variation, from styles of text mining pursued for purposes wherein time does not necessarily factor, for instance modeling fundamental differences between different languages and linguistic groups. Insights derived from historical analysis are relevant to scholars from history, art history, literature, political science, and economics, as well as practitioners at the law and business schools. Robust historical analysis is almost always improved by the insights of history scholars. The phrase “text mining for historical analysis” thus invokes a wide zone where text mining is used for pragmatic reasons.

中文译文

" 历史分析 " 标记了对不同时间文本的挖掘，该挖掘对所有不同时间经验的建模进行了培训，从为时间不一定考虑的目的而进行文本挖掘的样式，例如模拟不同语言和语言群体之间的根本差异。 从历史分析中得出的见解与历史，艺术史，文学，政治学，经济学的学者以及法学院和商学院的从业者相关。强有力的历史分析几乎总是通过历史学者的洞察力来改进。因此，“用于历史分析的文本挖掘”这一短语引用了一个宽阔的地带，其中出于实际原因使用了文本挖掘。

ENGLISH ORIGINAL

Unlike the phrase “digital history,” text mining for historical analysis carries no suggestion that computer-driven history will somehow substitute for history by traditional means - a threat that some members of the Temporal Cartographers’ Society are all too wary of. Text mining for historical analysis rather sits across the many domains that use text, AI and statistics to produce certain kinds of knowledge, useful to certain purposes. The hybrid thinking that has emerged in between quantitative and qualitative disciplines combines insights from history with those from information science has the power to catalyze a revolution in smart data. It is hybrid rather than merely interdisciplinary because what is produced in the process is more than an adjustment to old conversations.

中文译文

与 " 数字历史 " 不同的是，用于历史分析的文本挖掘没有暗示计算机驱动的历史会以传统手段以某种方式取代历史。社会太谨慎了 用于历史分析的文本挖掘是跨越许多领域，这些领域利用文本、AI和统计来产生某些类型的知识，对某些目的有用。 在定量和定性学科之间出现的混合思维将历史的洞察力与信息科学的洞察力结合起来，具有催化智能数据革命的力量。它是杂交的，而不仅仅是跨学科的，因为这一过程所产生的不仅仅是对旧对话的调整。

ENGLISH ORIGINAL

Something genuinely new is being produced where old fields meet, and this hybrid practice has the power to produce insight about large-scale troves of texts, not unlike the hybrid ears of corn that revolutionized industrial agriculture in the middle of the twentieth century. 18 Such a practice as this has a great deal to offer the community of scholars, although it is hard to see text mining replacing the many forms of work with data, documents, and archives that mark the specialty of history. The greatest beneficiaries of a hybrid practice of historical thinking with data are outside the walls of the academy.

中文译文

在旧田地交汇的地方正在产生一些真正新的东西，这种杂交做法能够产生对大量文字的洞察力，这与20世纪中叶使工业农业革命的玉米杂交耳不同. 18 这种做法对学者群体有很大帮助，尽管很难看到文本挖掘用标志着历史特色的数据、文件和档案来取代多种形式的工作。 历史思维与数据混合做法的最大受益者在学院的墙外。

Toward a Smarter Data Science

迈向更智慧的数据科学

ENGLISH ORIGINAL

Data science today needs to become smarter in ways that benefit from learning that originates outside the study of data. In Chapter 1, I tell the story of how my students plotted the trend of mentions of “ignorant women” over the course of debates from Britain’s parliament in the nineteenth century. They concluded that women had become more and more ignorant over the course of the century - rather than reasoning as a historian, that Britain’s parliament had become more and more biased against women over the course of the same time. I learned about the dangers of text mining because I welcome students of engineering, statistics, and business, as well as history and literature majors, to my classroom. The gap in the education of a significant proportion of my students is a warning sign. If some of my cleverest students are making those mistakes today, then the titans of industry will be making similar mistakes tomorrow.

中文译文

今天的数据科学需要变得更加聪明，从研究数据以外的学习中受益。 在第一章中，我讲述了我的学生在十九世纪英国议会辩论期间如何策划提及 " 侮辱妇女 " 的趋势。 他们得出结论，在本世纪中，妇女越来越无知，而不是作为历史学家的推理，英国议会在同一时期对妇女越来越有偏见。 我了解了文本挖掘的危险，因为我欢迎工程、统计和商业以及历史和文学专业的学生来到我的教室。 相当一部分学生的教育差距是一个警示信号。如果一些我最聪明的学生今天犯这些错误，那么工业的巨人明天也会犯类似的错误。

ENGLISH ORIGINAL

The solution to bad graphs and foolish interpretations is, I believe, a return to reality. Reality, in terms of our collective experience with each other, is perhaps only known through stories about our past, and there are many other kinds of stories that only speak to desire, its reflections, and shadows. This is why hybrid knowledge matters. It’s why any university that’s teaching data science urgently needs to plan programs of instruction that foreground not only the skills of working with data at speed but also understanding what data means. We can use data to discover important trends, recent and older, that give us insights to the subtle changes that are shifting society. But to use those tools effectively, we need smart data and hybrid knowledge, attuned to the latent bias of the data and of algorithm and how to navigate them. Data-driven analysis of language, without a historical method, is apt to produce unusable information.

中文译文

我认为，解决糟糕的图表和愚蠢的解释是回到现实。 从我们彼此的集体经历来看，现实也许只能通过我们过去的故事来认识，还有许多其他类型的故事，它们只讲欲望，它的反射，和阴影。这就是为什么混合知识很重要。 任何教授数据科学的大学都急需规划教学计划，我们可以利用数据来发现重要的趋势，无论是最近还是更老的趋势，使我们能洞察到正在改变的社会的微妙变化。 但是要有效地使用这些工具，我们需要智能数据和混合知识，适应数据和算法的潜在偏差，以及如何导航。 数据驱动对语言的分析，没有历史方法，就适宜产生无法使用的信息.

ENGLISH ORIGINAL

Without help from abroad, the world without history is in trouble. In spaces like digital history where computationalists and historians meet and ask questions about each other’s basic premises, problems tend to emerge that were previously unknown to either field beforehand. When historians consult with algorithms, for example, they tend to move deliberately, reflecting on the fit of each algorithm for each separate stage of analysis. They care about distinguishing the signals of correlation from those of causation. These conversations can be productive of new insight and opportunities for rigor. A truly hybrid research process forces colleagues to submit their unstated hypotheses to careful testing. It requires refraining from hasty diagnoses and cheap answers until the process for arriving at truth meets the standards of fields that have never previously reviewed their terms of agreement.

中文译文

没有国外的帮助，没有历史的世界就陷入困境。在像数字历史这样的空间中，计算学家和历史学家会聚一堂，互相询问彼此的基本前提，问题往往会出现，而这两个领域之前都不了解。 例如，历史学家咨询算法时，往往会刻意移动，反思每个分析阶段的每个算法是否合适。他们关心区分相关信号和因果关系信号。 这些对话可以产生新的洞察力和严格度的机会。 一个真正的混合研究过程迫使同事将其未陈述的假设提交仔细测试。这需要避免仓促诊断和廉价答案，直到查明真相的过程达到以前从未审查过协议条款的领域标准。

ENGLISH ORIGINAL

The birth of new fields where old disciplines meet is a reality of our time. Data science is sometimes presented, in its ideal form, as the application of statistics to big data with substantive knowledge. 19 For this reason, teams of medical researchers work alongside statisticians and memory specialists to produce the field of biostatistics; teams of linguists collaborate with computer scientists and statisticians to produce computational linguistics; and political scientists and social scientists work with technology experts to produce computational social science. Now, it seems clear, a new field is emerging that engages text mining with the aim of establishing historical truth from the humanities and history in particular. Hybrid thinking about historical truth has much to offer the rest of the world. It can aid political journalists who are constantly seeking new ways of understanding cultural differences in the electorate.

中文译文

旧学科交汇的新领域的诞生，是我们时代的现实。数据科学有时以理想的形式被表述为统计对具有实质性知识的大数据的应用. 19 为此原因，医疗研究人员团队与统计人员和记忆专家一起工作，生产生物统计学领域；语言学家团队与计算机科学家和统计人员合作，生产计算语言学；政治科学家和社会科学家与技术专家合作，生产计算社会科学。 现在，似乎很明显，一个新的领域正在出现，它涉及文本挖掘，目的是从人文和特别是历史中确定历史真相。 对历史真理的混合思考为世界其他地方提供了很多东西。 它可以帮助政治记者不断寻求新的方法来理解选民的文化差异。

ENGLISH ORIGINAL

It can support media streaming companies that wish to learn more about who identifies with certain kinds of music, lyrics, and narratives, and what kinds of content leaves them cold. These practices can entertain and educate the general public, which is fascinated by data visualizations that show them aspects of their own experience from unexpected angles. And it can support scholars already dedicated to a careful and critical analysis of who human beings are, what was changing at any given movement in the past, and where culture appears to be heading. Yet most of the tools from data science today have been developed without consultation with ideas about truth from history or the humanities. Like the wordcloud, they are simplistic approximations of truth where information is lost.

中文译文

它可以支持那些希望更多地了解谁认同某些种类的音乐，歌词，和叙事的媒体流媒体公司，以及哪些内容令它们变得冷淡。这些做法可以娱乐和教育大众，大众对数据可视化感兴趣，从意想不到的角度来显示自己经验的方面。它可以支持已经致力于 仔细和批判性分析的学者们，分析人类是谁，分析过去任何运动的变化，分析文化的走向。 然而，今天数据科学的大多数工具是在没有与历史或人文学科的真理思想协商的情况下开发的。 与词云一样，它们也是信息丢失之处真理的简单近似.

ENGLISH ORIGINAL

Mathematician Cathy O’Neil has documented how bail extension rulings by judge became increasingly racist after the judges were advised by an AI tool whose advice was based on data from racist moment in the past that used data from previous decades. Bias riddles contemporary AI products today. 20 Even GPT-3 cannot be trusted to summarize even relatively simple prose without the service of a trained fact checker. 21 The algorithms designed to correct for bias, meanwhile, rarely go to the root of the problem, because most of them were designed by information scientists who lack a familiarity with how the bias got into the data in the first place. Today’s AI is simply not capable of performing at the rigorous standard of truth that scholars in the liberal arts have long expected of each other. The cost to society of data science ignoring the liberal arts is enormous.

中文译文

数学家凯茜· 奥尼尔(Cathy O ' Neil)记录了法官通过AI工具向法官提供建议后，保释延期裁决如何变得日益种族主义。 今天的比亚斯谜语是当代AI产品. 20 即使是GPT-3,也不可能在没有受过训练的事实检查员服务的情况下总结即使是相对简单的散文. 21 同时，为纠正偏差而设计的算法很少触及问题的根源，因为这些算法大多是信息科学家设计的，他们最初并不熟悉偏差是如何进入数据的。今天的AI根本无法以人文学科界学者长期期待的严格真理标准来表演。无视人文学科的数据科学对社会的代价是巨大的.

ENGLISH ORIGINAL

It is represented in quantifiable terms by the lakes of unused textual data that cannot be effectively exploited by today's AI tools. Statisticians estimate that, in 2017, phone users around the world sent 6.8 billion text messages a day. 22 In 2018, the world's population was 86.5 percent literate, and that literate bulk of humanity composed roughly 281 billion emails per day. 23 Indeed, our civilization produces a shocking amount of text every year. Since the advent of electronic systems, such as word processors and webpages, for authoring and sharing text, the amount of text created and preserved in analyzable form has dwarfed earlier numbers.

中文译文

以数量表示的是，今天的人工智能工具无法有效地利用未使用的文字数据湖。 统计学家估计，2017年，世界各地的电话用户每天发送68亿条短信. 22 2018年，世界人口识字率为86.5%,而识字的人类大部分每天大约为2810亿电子邮件。 23 事实上，我们的文明每年产生令人震惊的文本。 自用于编写和共享文本的电子系统，如文字处理器和网页问世以来，以可分析形式创建和保存的文本数量已比早先的数字相形见绌。

ENGLISH ORIGINAL

If printed out, the text on the internet itself would consume 136.2 billion pieces of paper. 24 Grand as these figures may be, they are modest compared with the wealth of disposable text that humans generate in the course of continuously communicating, redacting, and recirculating information. 25 One approximation of that number, however, is that the US federal government employees print out about 18 billion pages of documents every year. "Big data just gets bigger and bigger and bigger," wrote Aiden and Michel, reflecting problems of scale from 2013.26 For want of effective text mining strategies, an ocean of data has gone virtually untouched. The want of interdisciplinary approaches to text mining that span humanistic questions and quantitative approaches also limits what industry can do with the data it already has.

中文译文

如果印刷出来，互联网上的文本本身将消耗1362亿张纸。 24 尽管这些数字可能很大，但与人类在不断沟通、编辑和重播信息过程中产生的大量可支配文本相比，它们并不多见。 25 然而，该数字的一个近似数字是，美国联邦政府雇员每年印制约180亿页的文件。 Aiden和Michel写道:“大数据正变得越来越大，由于缺乏有效的文本挖掘战略，数据大海几乎没有受到影响。 跨人文问题和定量方法的跨学科文本挖掘方法的缺乏，也限制了业界对其已有数据所能做的.

ENGLISH ORIGINAL

During a conversation with an executive at a streaming movie service, I recently had the occasion to explain the recent successes that define the modern field of text mining in the liberal arts. Curious about my projects, he asked what text mining could do for his own industry and I told him about a dozen new studies from the digital humanities. I explained how scholars in the digital humanities had been counting the underrepresentation of black actors in Hollywood scripts, and observing the structural differences between novels written by women and those written by men. Because the executive's job was to buy and sell scripts that target the preferences of a wide range of viewers, I offered speculation about how text mining might affect his industry.

中文译文

最近我有机会解释在人文学科界现代文本挖掘领域最近的成功。 他想知道我的项目，他问，文本挖掘能为他自己的产业做什么，我告诉他十几个新的研究 从数字人文。 我解释了数字人文的学者们是如何统计好莱坞剧本中黑人演员人数不足的，并观察了女性写作的小说和男性写作小说之间的结构差异。因为执行官的工作是买卖针对广大观众喜好的剧本，所以我提出了关于文本挖掘会如何影响他的行业的猜测.

ENGLISH ORIGINAL

I imagined that he might do well to hire a digital humanist as these scholars, in recent years, have been modeling the imaginary dimensions of identity and plot, and creating variables that could be measured against the voluminous data the executive had been gathering. While the company already knew a great deal about which movies were hits and which were duds, a digital humanist could create a model based upon conceptual categories that would explain their viewing behavior and thus offer them movies that best fit their needs. If women stop watching twenty minutes in, for example, near a scene when the men are doing all the talking, a humanist could train an algorithm to recognize the common pattern and its effects on viewing patterns. The executive agreed. He was thoroughly hooked on the algorithms I described, and he wanted to add them to his business model.

中文译文

我想象着他雇用一位数字人文主义者可能做得好，因为近年来这些学者一直在模拟身份和情节的假想维度，并创造出可以根据行政主管收集的大量数据来衡量的变量。 虽然公司已经非常了解哪些电影是点击的，哪些是哑剧，但数字人文主义者可以创建基于概念分类的模型来解释他们的观看行为，从而为他们提供最适合他们需要的电影。如果女性停止观看20分钟，比如说，在男性进行所有交谈的场景附近，一个人文主义者可以训练一个算法来识别共同的图案及其对观看图案的影响。行政部门同意了。 他完全迷上了我描述的算法 他想把它们添加到他的商业模型中

ENGLISH ORIGINAL

But here’s the rub, he confessed: he would never be able to sell his bosses on hiring a PhD in literature - a field that has produced some of the finest scholarship on text mining applied to culture. The prejudice against the humanities in Silicon Valley went too deep, he explained. Creating smart data requires becoming smart about data; and if we want the data to provide the most complete reflection of human experience possible, we must try to make a happy marriage between traditional tools of reasoning that have been honed over centuries, and contemporary practices with data. Having shunned an active collaboration with humanists, Silicon Valley is not nearly as smart as it could be. Big tech needs an engaged, technical conversation about radically improving the way we work with data to model human experience.

中文译文

他承认:他不可能在聘请文学博士时出卖老板，他解释说，对硅谷人文科学的偏见太深。 创造智能数据需要变得对数据很聪明；如果我们希望数据能提供尽可能完整的人类经验的反映，那么我们必须努力在经过几百年磨练的传统推理工具与现代的数据实践之间实现幸福的结合。 在回避了与人文主义者的积极合作之后，硅谷并没有那么聪明。大科技需要参与，技术对话 关于彻底改善我们的工作方式 用数据模拟人类的经验。

ENGLISH ORIGINAL

Many civilizations achieved mass literacy sometime in the nineteenth century; but that didn’t mean that every reader could participate in a conversation about hermeneutics. Silicon Valley hires hundreds of computer scientists with basic training in natural language processing, but that doesn’t mean they understand what a critical engagement with human experience looks like. If the culture of Silicon Valley added teams of professional humanists to its teams that work on data analysis, this marriage could give birth to a data-driven revolution far more subtle, and far more accurate, than either camp could manage on their own. For such a thing to happen, however, hybrid thinking across data science and the liberal arts must become routine in the places that drive research innovation - in the universities themselves.

中文译文

许多文明在十九世纪的某个时候实现了大众识字；但这并不意味着每个读者都可以参与关于美食的谈话。 硅谷雇佣数百名计算机科学家，接受自然语言处理的基本训练，如果硅谷文化将专业人文主义者的团队加入到其从事数据分析的团队中，那么这段婚姻可以产生一场由数据驱动的革命，比两个阵营都能够自己管理得更微妙，更准确得多。然而，要做到这一点，数据科学与人文学科之间的混合思维必须成为推动研究创新的地方——大学本身——的常规。

ENGLISH ORIGINAL

The $ 100 million gifts that have created new schools of information at Cornell University, Harvard, MIT, Stanford, and Johns Hopkins have so far made room for very few scholars with a liberal arts background. The exclusion of the humanities from data science centers up to this point is understandable. It is, after all, only in the last few years that a real watershed has been crossed where the contribution of historical thinking to text mining is visible. Dozens if not hundreds of colleagues have contributed to that effort, some of them members of the Temporal Cartography Society who work with text on paper in dusty archives, and some of them former members who ventured into the no man's land and there met up with a few brave data scientists from the world without history.

中文译文

在康奈尔大学、哈佛大学、麻省理工学院、斯坦福大学和约翰· 霍普金斯创立了新的信息学院的1亿美元的礼物，迄今为极少数具有人文学科背景的学者提供了空间。 迄今将人文科学排除在数据科学中心之外是可以理解的。 毕竟，只是在过去几年中，才跨越了一个真正的分水岭，历史思维对文本挖掘的贡献是显而易见的。 数十位甚至数百位同事为这一努力做出了贡献，其中一些是临时制图学会的成员，他们用纸上写在灰尘档案中，其中一些是前成员，他们冒险进入无人之地，与来自世界上一些没有历史的勇敢的数据科学家会面。

ENGLISH ORIGINAL

My modest contribution is to assemble the discoveries of others, alongside a few of my own, into a teachable series of problems about how historical questions can reshape data science into a smarter discipline.

中文译文

我微薄的贡献是把其他人的发现，和我的几个人一起，汇集成一系列可以教化的问题，关于历史问题如何将数据科学重塑成更聪明的学科。

About the Chapters

各章简介

ENGLISH ORIGINAL

Part I of this book opens with a catalog of dangers about where contemporary data practices are failing. Drawing on a decade of interdisciplinary research, it lays out best practices of how liberal arts approaches (especially those from history) can make work with textual data more robust. Chapter 1 tours a series of high-stakes disasters from data science, where papers retracted after publication have demonstrated how riddled contemporary data-driven practices are with error. It categorizes the major typology of dangers that lead analysts into trouble, drawing especial attention to insufficient engagement with historical context and fantasies of predictive knowledge that cause analysts to rush toward law-like conclusions without stopping to theorize the work algorithms must do to uncover the logic of the past.

中文译文

本书第一部分以关于当代数据实践失败之处的危险目录开始。 它借鉴了十年的跨学科研究，阐述了人文学科方法(特别是来自历史的艺术方法)如何使文本数据工作更加有力的最佳做法。 第一章从数据科学中游览了一系列重大灾害，出版后收回的论文表明，当代数据驱动的做法有误。 它将导致分析人员陷入麻烦的主要危险类型分类，特别提请注意与历史语境的接触不足以及预测性知识的幻想，这些预测性知识使分析人员急于作出类似法律的结论，而不停止理论工作算法必须做的揭示过去逻辑的工作。

ENGLISH ORIGINAL

Chapter 2 proposes a series of remedies for poor thinking with data, especially care about the provenance of data and an interest in the fundamentals of historical thinking can guide data-driven research processes. The chapter introduces several teams of interdisciplinary analysts whose work forms the inspiration for this book. Their ranks include traditional scholars from the liberal arts who collaborate with information scientists as well as scholars like myself who have trained in liberal arts subjects but sometimes work with data and code. Chapter 3 dives deeper into the problem of treating words with care, explaining how words can act as barriers to interpretive understanding but also keys that unlock powerful analyses of the past. Chapter 4 extrapolates a theory of a research process capable of working between qualitative questions and quantitative tools.

中文译文

第2章提出了一系列对用数据思考不良的补救办法，特别是注意数据的来源，对历史思考基本原理的兴趣可以指导数据驱动的研究过程。本章介绍了若干跨学科分析员小组，他们的工作为这本书提供了灵感。 他们的队伍包括来自人文学科的传统学者，他们与信息科学家合作，以及像我这样的学者，他们受过人文学科科目的培训，但有时还从事数据和代码方面的工作。第3章更深入地探讨了谨慎对待词的问题，解释了词如何能够成为解释理解的障碍，同时也是解开过去强大分析的钥匙。第四章推断出一种研究过程的理论，这种理论能够在定性问题和定量工具之间发挥作用.

ENGLISH ORIGINAL

It proposes a series of steps for how a “critical” encounter with a digital archive and quantitative algorithm might proceed in such a way as to preserve humanistic concerns with meaning, bias, silence, and perspective. The result is a “critical search” for the past, which can form part of a larger agenda of re-search, but which is fundamentally better suited to humanistic questions than searches conducted with out-of-the-box tools like wordclouds. Chapter 5 steps back to the purposes that data scientists sometimes bring to historical data, investigating one of the chief subjects of misunderstanding and conflict between quantitative and qualitative approaches. It asks head-on whether excavations into the past can support a “predictive” theory of the future. Reviewing historians’ positions on this subject reveals both numerous objections to predictive thinking with historical data as well as a handful of important opportunities.

中文译文

它提出了一系列步骤，说明如何在数字档案和定量算法的“关键”交锋中，用意义、偏见、沉默和视角保护人文主义关切。 结果是对过去的“批判性搜索”,这可以构成更大范围的再研究议程的一部分，但从根本上来说，它更适合人文主义问题，而不是使用诸如文词云之类的工具进行搜索。 第5章回到数据科学家有时给历史数据带来的目的，调查在定量和定性方法之间产生误解和失节的一个主要主题。 它直面地询问，挖掘过去是否支持对未来的“预测”理论。 历史学家对这个主题的立场审查表明，许多人反对用历史数据来预测思想，也有一些重要的机会。

ENGLISH ORIGINAL

Part II of this book explores how ideas from the philosophy of history can help us to study historical change. Drawing on the scholarship of many contemporary philosophers and historians who have wrestled with the problem of how we know about the past, this Part argues that the past is a multidimensional object, in that long-term forces and short-term shifts, experience and memory, all work together to shape the past, their interactions constantly changing. Chapter 6 draws on the philosophy of history to propose a series of categories of temporal experience appropriate as subjects for data-driven modeling. Next, we consider question of whether mathematics, data, and modeling can expand the analyst’s ability to apprehend the multiple dimensions of change in the historic past. To that purpose, I present a concrete series of case studies where I engage datasets around a theory of historical change.

中文译文

本书第二部分探讨了历史哲学中的思想如何帮助我们研究历史变化。本部分借鉴了许多当代哲学家和历史学家的奖学金，他们奋力解决我们如何认识过去的问题，认为过去是一个多层面的目标，在这种长期的力量和短期的转变，经验和记忆中，都共同努力塑造过去，他们的互动不断在变化。第6章借鉴历史哲学，提出一系列适合作为数据驱动模型主体的时间经验类别。其次，我们考虑的是，数学、数据和建模能否扩大分析师的能力，以抓住历史中的变化的多重层面。 为此，我提出一系列具体的个案研究，围绕历史变化的理论使用数据集。

ENGLISH ORIGINAL

In Chapter 7, I deploy a series of text-mining methods to study how the process of memory changed in nineteenth-century Britain. I present multiple possible quantitative approaches that open up different dimensions of the past, and I show how each approach, and each subinvestigation within each approach, adds nuance and complication to our understanding of the past. Chapter 8 discusses algorithms that can be used to detect the distinctiveness of different eras, and explores how those algorithms can be adjusted to highlight the uniqueness of experiences that took place over a day, week, year, or decade. Chapter 9 uses data to study how individuals influence each other.

中文译文

在第七章中，我运用了一系列的文本挖掘方法来研究十九世纪英国记忆过程的变化。我提出多种可能的量化方法，这些方法打开了过去最模糊的维度，我说明每一种方法如何增加细微差别和复杂程度，以及每一种方法内的每一次调查如何增加我们对过去的理解。 第八章讨论可用于检测不同时代独特性的算法，并探索如何调整这些算法，以突出在一天，一周，一年，十年间发生的经历的独特性。第9章使用数据来研究个人如何相互流利.

ENGLISH ORIGINAL

In Chapter 10, I show that algorithms can be used to investigate slow-moving and quick-moving pulses of temporal change at any moment in the past; they can also identify the trends that were at the end of their life, fossils that represented a temporary fad, and other shapes of time that have traditionally been identified through qualitative means. I also theorize what it means to critically adjust an algorithm to a historical question. Chapter 11 introduces theories of “modernization” and suggests several algorithmic strategies for looking at long-term development. At the same time, this chapter proposes important limits to the use of data to revisit the most important questions about modernity, poverty, and marginalization that historians have treated in other ways. In Chapter 12, I investigate a concrete historical question: when and how attacks on environmentalists in the US Congress delayed action on climate change.

中文译文

在第十章中，我表明算法可以用来调查过去任何时刻时间变化的缓慢移动和快速移动的脉冲；它们也可以确定生命末期的趋势，代表暂时快感的化石，以及传统上通过定性手段确定的其他时间形状。我也认为 严格调整算法 适应历史问题意味着什么 第11章介绍了“现代化”理论，并提出了若干着眼于长期发展的算法战略。 同时，本章还提出了对使用数据重温历史学家以其他方式处理的关于现代化，贫困和边缘化等最重要问题的重要限制。我在第12章中调查了一个具体的历史问题:美国国会对环保主义者的攻击何时以及如何拖延了对气候变化的行动。

ENGLISH ORIGINAL

It argues against the testing of hypotheses in the style of many machine-learning papers, drawing on historical argumentation to propose a process where meaningful insight about the past is the standard of discovery. I also demonstrate, to a degree lacking in the other chapters, what it looks like to work between a data visualization and in-text mentions. Alternating data-driven work with the traditional methods of history makes for a more accurate, nuanced, and robust practice, even when the results are circulated in the form of a data-driven visualization. In the data-driven case studies of Part II, AI and statistics are a subject for critical inquiry. Of each algorithm, we will ask: Does this approach get us closer to real knowledge, as understood by contemporary historical theories of how memory, temporality, or causality work?

中文译文

它反对在许多机器学习论文的风格中测试假说，利用历史论证提出一个过程，让对过去有意义的见解成为发现的标准。我还以其他章节所缺乏的程度，表明在数据可视化和内文提及之间工作的样子。 将数据驱动工作与历史传统方法相结合，使得实践更加准确，细微，有力，即使结果以数据驱动可视化的形式流传。在第二部分数据驱动的案例研究中，AI和统计是关键调查的主题。在每一种算法中，我们将问:这个方法是否使我们更接近于真实的知识，正如当代历史理论对记忆、时间或因果关系如何起作用的理解?

ENGLISH ORIGINAL

How might small adjustments in the parameters of the algorithm or the controlled vocabulary produce different results? The individual chapters in the Part II will introduce algorithms from vintage statistics such as divergence, modern AI approaches to clustering such as topic modelling, and machine-learning dependencies such as word embeddings. Nonspecialists need not shut the book in alarm; each algorithm will be introduced in accessible language, and the focus of each chapter will remain on the larger question of how hybrid scholarship requires critical thinking about the fit between data, algorithms, and models of historical change. Some caveats are important. While certain historical “discoveries” are revealed in the middle chapters of this book, I am not primarily concerned in this book with incontrovertible historical argumentation, but rather with a practice designed to combine historical insight with algorithmic rigor.

中文译文

算法参数或受控词汇的小调整如何产生不同的结果? 第二部分的个别章节将引入来自差异等古代统计的算法，对主题建模等集群的现代AI方法，对词嵌入等机器学习依赖性。非专家不需要惊慌地关闭本书；每个算法都会被引入无障碍语言，每一章的重点仍将是混合奖学金需要怎样批判性地思考数据，算法，以及历史变化模型之间的适合性这个更大的问题。一些警告很重要。 虽然这本书的中间章节中揭示了某些历史的"发现",但我在本书中并不主要关注无可争议的历史争论，而是关注一种旨在将历史洞察力与算法刚性结合起来的做法.

ENGLISH ORIGINAL

Many of the “discoveries” of data science in recent years merely reproduce standard knowledge; to this end, I am clear where my exercises in data science do the same, although my case studies generally aim for a higher standard of insight. I also emphasize where inadequate data means that the data-driven discoveries in question need to be traced back to archives for the historical explanation to be complete. Chapter 12 looks in greater detail into the notion of “explanation” in history, and how it is that data-driven discoveries can rise to the high standard that historians hold for actual discovery. It is an axiom of the method presented here that data-driven analysis is insu fficient for many historical problems. A savvy analyst must apply her skills of skepticism to each phenomenon apparent in the data, validating each discovery by work with the text.

中文译文

近年来许多数据科学的“发现”只是复制了标准知识；为此，我很清楚我在数据科学中的练习在哪里做同样的工作，尽管我的案例研究一般是为了提高洞察力的标准。 我还强调，在哪些地方，数据不足意味着需要将有关数据驱动的发现追溯到档案室，以便完成历史解释。 第12章更详细地审视了历史上的“解释”概念，以及数据驱动的发现如何能够上升到历史学家认为实际发现的高标准。 数据驱动分析不适应许多历史问题，这是本文所述方法的一个定理。 经验丰富的分析师必须将她的怀疑技巧应用于数据中显而易见的每个现象，通过与文本合作验证每个发现.

ENGLISH ORIGINAL

Only then can data science rise to the ideal of complete explanation to which historians aspire. Finally, Part III steps back from detailed engagements with methods to ask what can be accomplished when critical thinking is brought to methods. Chapter 13 looks at how data helped historical linguists to purge centuries of racial bias from their map of the relationships between world languages. It takes a long-term view on the development of disciplines to imagine what our age of reform and data might contribute to other foundational systems of understanding in the humanities and social sciences. Chapter 14 investigates the future of research. It articulates a call for philosophers of history to help us to understand more of the dimensions of the past and for data scientists to use historical understanding to investigate the limits and promise of their models.

中文译文

只有这样，数据科学才能上升到历史学家所渴望的完全解释的理想。最后，第三部分从详细接触的角度出发，提出一些方法，询问在将批判性思维引入方法时可以取得什么成就。 第13章审视了数据如何帮助历史语言学家从他们对世界语言之间关系的地图中清除几个世纪的种族偏见。它从长远的角度看待学科的发展，来想象我们改革的时代和数据对人文和社会科学中其他基本的理解体系可能有什么贡献。第14章调查研究的未来。它明确呼吁历史哲学家帮助我们更多地了解过去的维度，并呼吁数据科学家利用历史理解来调查其模型的局限性和前景.

ENGLISH ORIGINAL

It envisions a robust future for practices of AI and data science in the future, one where (some) humanists and (certain) historians have an important role to play. It calls on us all to contribute to the collective understanding of where text mining can answer big questions and where traditional research is warranted. Finally, it glances at some public uses of text mining for the purpose of transparency and political reform.

中文译文

它设想了未来AI和数据科学实践的强劲未来，其中(某些)人文主义者和(某些)历史学家可以发挥重要作用。 它呼吁我们大家促进集体理解文本挖掘在哪里能够回答大问题，在哪里应当进行传统研究。 最后，报告审视了为了透明度和政治改革的目的公开使用文本挖掘的一些情况。

ENGLISH ORIGINAL

Ultimately, most humanists are already concerned with evidence and analysis. Understanding the past with accuracy and precision is a matter of evidence, and the humanities share this concern for reality with the sciences. Reading the speeches of presidents, comparing novels, studying paintings, or parsing old maps give us evidence, and many humanists pride themselves on their superior talents of attention, comparison, and interpretation. Text mining can benefit from humanists' careful discernment about the dangers of biased data and the promise of careful categorization of experience. Like other humanistic enterprises, text mining is therefore best pursued with some knowledge of historical context, of the limits of each body of evidence, and the methodological questions suited to those objects in particular.

中文译文

归根结底，大多数人文学者本就关心证据与分析。准确、精确地理解过去，是证据问题；人文学科与科学一样重视现实。阅读总统演讲、比较小说、研究绘画或解析旧地图，都为我们提供证据；许多人文学者也以自己卓越的观察、比较和阐释能力为傲。文本挖掘可以受益于人文学者对偏见数据风险的谨慎辨识，以及他们对经验进行细致分类的能力。因此，与其他人文研究一样，开展文本挖掘时，最好了解历史语境、每一类证据的局限，以及最适合这些对象的方法论问题。

Notes

注释

ENGLISH ORIGINAL

1. Shadi Shahsavari, Pavan Holur Timothy R. Tangherlini, and Vwani Roychowdhury, "Conspiracy in the Time of Corona: Automatic Detection of Covid-19 Conspiracy Theories in Social Media and the News," ArXiv Preprint ArXiv: 2004.13783, 2020; Richard Jean So, Redlining Culture (New York: Columbia University Press, 2020);

中文译文

1. Shadi Shahsavari,Pavan Holur Timothy R. Tangherlini,和Vwani Roychowdhury,"科罗纳时代的阴谋:社会媒体和新闻中自动发现新冠病毒-19阴谋论",ArXiv Preprint ArXiv:2004.13783,2020年；Richard Jean So, Redlining Culture(纽约:哥伦比亚大学出版社，2020年);

ENGLISH ORIGINAL

2. I am not myself an American historian, but I record the relative surprise of my collaborator, Peter Kastor, with whom I am writing about word embeddings in Congress.

中文译文

2. 我不是美国历史学家，但我记录了我的合作者彼得· 卡斯特的相对惊讶，我正与他一起写关于国会中嵌入文字的文字。

ENGLISH ORIGINAL

3. "Inauguration Speeches in Wordclouds," The Guardian, January 21, 2013, sec. US News, www.theguardian.com/world/gallery/2009/jan/21/obama-inauguration-word-clouds-gallery.

中文译文

3. "Wordclouds的就职演说"，《卫报》,2013年1月21日，sec。 美国新闻，
www.theguardian.com/world/gallery/2009/jan/21/obama-inauguration-word-clouds-gallery.

ENGLISH ORIGINAL

4. Jessica Estepa, “House Democrat Makes a Word Cloud of Trump Insults,” USA Today, September 26, 2017, www.usatoday.com/story/news/politics/onpolitics/2017/09/26/house-democrat-makesword-cloud-trump-insults/705120001.

中文译文

4. 杰西卡· 埃斯特帕，"豪斯民主党制造特朗普侮辱的词云","今日美国",2017年9月26日 www.usatoday.com/story/news/politics/onpolitics/2017/09/26/house-democrat-makesword-cloud-trump-insults/705120001.

ENGLISH ORIGINAL

5. Ben Schott, “Biden and Trump Go Cloud to Cloud,” Bloomberg, August 28, 2020, www.bloomberg.com/opinion/articles/2020-08-28/biden-and-trump-nomination-speeches-are-two-clashing-wordclouds.

中文译文

5. Ben Schott, "Biden and Trump Go Cloud to Cloud," 彭博社，2020年8月28日， www.bloomberg.com/opinion/articles/2020-08-28/biden-and-trump-nomination-speeches-are-two-clashing-wordclouds.

ENGLISH ORIGINAL

6. Sewell, “Three Temporalities,” 245-80.

中文译文

6. Sewell, " 三个时间 " ,245-80。

ENGLISH ORIGINAL

7. Steven Ruggles, “The Revival of quantification: reflections on Old New Histories,” Social Science History 45:1 (October 1, 2021): 1-25, https://doi.org/10.1017/ssh.2020.44; Sam White, Christian Pfister, and Franz Mauelshagen, The Palgrave Handbook of Climate History (London: Palgrave, 2018).

中文译文

7. Steven Ruggles, " 量化的复兴:对旧新历史的反思，"社会科学史45:1(2021年10月1日): 1–25, https://doi.org/10.1017/ssh.2020.44; Sam White, Christian Pfister, and Franz Mauelshagen, The Palgrave Handbook of Climate History (London: Palgrave, 2018).

ENGLISH ORIGINAL

8. Kate Crawford, The Atlas of AI: Power, Politics, and the Planetary Costs of artificial Intelligence (New Haven, CT: Yale University Press, 2021).

中文译文

8. Kate Crawford, The Atlas of AI: Power, Politics, 以及人工智能的行星成本(New Haven, CT: 耶鲁大学出版社，2021).

ENGLISH ORIGINAL

9. For the Philadelphia Project, see Luke Blaxill, The War of Words: The Language of British Elections, 1880-1914, Royal Historical Society, Studies in History. New Series 103 (Woodbridge; Rochester, NY: The Boydell Press, 2020); Robert Fogel and Stanley Engerman, Time on the Cross (Boston, MA:Little, Brown and Company, 1974).

中文译文

9. 关于费城计划，见Luke Blaxill,《文字战争:英国选举语言》,1880-1914,皇家历史学会，历史研究。 新系列103 (Woodbridge; Rochester, NY: The Boydell Press, 2020); 罗伯特· 福格尔和斯坦利· 恩格曼，Time on the Cross (Boston, MA: Little, Brown and Company, 1974).

ENGLISH ORIGINAL

10. Blaxill, The War of Words, 22 n. 9.

中文译文

10. Blaxill,The War of Words, 22 n. 9 (英语).

ENGLISH ORIGINAL

11. For the comparative evolution of political science and history and broader meditation on quantitative methods suited to data, see Gregory Wawro and Ira Katznelson, Time Counts: Quantitative Analysis for Historical Social Science (Princeton, NJ: Princeton University Press, 2022). I draw on innovative work from political science and sociology throughout this book, but two books stand out as recent statements about methods in the field: Brandon Stewart, Justin Grimmer, and Margaret E. Roberts, Text as Data (Princeton, NJ: Princeton University Press, 2022), and Xiaoling Shu, Knowledge Discovery in the Social Sciences: A Data Mining Approach (Berkeley, CA: University of California, 2020).

中文译文

11. 关于政治学和历史的比较演变，以及对适合数据的定量方法的更广泛的冥想，见Gregory Wawro和Ira Katznelson，《时间统计:历史社会科学的定量分析》(Princeton,NJ:普林斯顿大学出版社，2022年)。 我在整个书中借鉴了政治学和社会学的创新著作，但有两本书作为关于该领域方法的最新声明而突出:布兰登· 斯图尔特，贾斯汀· 格里默，和玛格丽特· E· 罗伯茨，Text as Data(Princeton, NJ:普林斯顿大学出版社，2022年),以及小林舒，"社会科学的知识发现:数据开采方法"(Berkeley, CA: University of California).

ENGLISH ORIGINAL

12. Blaxill, The War of Words.

中文译文

12. Blaxill,"言语之战".

ENGLISH ORIGINAL

13. Jessica Marie Johnson, “Markup Bodies: Black [Life] Studies and Slavery [Death] Studies at the Digital Crossroads,” Social Text 36:4(137) (December 1, 2018): 57-79, https://doi.org/10.1215/01642472-7145658; Roopika Risam, New Digital Worlds: Postcolonial Digital Humanities in Theory, Praxis, and Pedagogy (Evanston, IL: Northwestern University Press, 2019), www.jstor.org/stable/10.2307/j.ctv7tq4hg.

中文译文

13. Jessica Marie Johnson, " Markup Bodys: Black [生命]研究与奴隶制[死亡]研究，数字十字路口，"社会文本36:4(137) (2018年12月1日): 57-79, https://doi.org/10.1215/01642472-7145658; Roopika Risam, 新数字世界: 后殖民时代数字人文理论，普拉希斯，和教育学(Evanston, IL:西北大学出版社，2019年), www.jstor.org/stable/10.2307/j.ctv7tq4hg.

ENGLISH ORIGINAL

14. The field of digital history has only just emerged as a proper field, with its first monograph in text mining (Blaxill, The War of Words) and the first monograph on using GIS applied to a historical question (Cameron Blevins, Paper Trails: The U.S. Post and the Making of the American West (New York: Oxford University Press, 2021).

中文译文

14. 数字历史领域刚刚作为一个适当的领域出现，其第一部专著是文本挖掘(Blaxill, The War of Words),第一部关于使用地理信息系统的专著适用于一个历史问题(Cameron Blevins, Paper Trails: The United States Post and the Making of the American West(纽约:牛津大学出版社，2021年)。

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15. Jo Guldi, “Parliament’s Debates about Infrastructure: An Exercise in Using Dynamic Topic Models to Synthesize Historical Change,” Technology and Culture 60 1 (March 21, 2019): 1-33; Jo Guldi, “The Official Mind’s View of Empire, in Miniature: Quantifying World Geography in Hansard’s Parliamentary Debates,” Journal of World History 32:2 (2021): 345-70, https://doi.org/10.1353/jwh.2021.0028; Jo Guldi, “The Algorithm: Mapping Long-Term Trends and Short-Term Change at Multiple Scales of Time,” The American Historical Review 127:2 (June 1, 2022): 895-911,

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15. Jo Guldi, " 议会关于基础设施的辩论:利用动态专题模型合成历史变化的实践 "，《技术和文化》,60 1 (2019年3月21日): 1–33; 乔· 古尔迪，"官方明德对帝国的看法，在迷你中:量化汉萨德议会辩论中的世界地理","世界历史杂志"32:2 (2021):345–70, https://doi.org/10.1353/jwh.2021.0028; 乔· 古尔迪，"算法:绘制长期趋势和短期变化的多尺度时间，"美国历史评论127:2(2022年6月1日):895–911,

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16. Dzovinar Kévonian and Philippe Rygiel, “Connected Ogres: Global Sources in the Digital Era,” Monde(s) 21:1 (May 11, 2022): 73-79; Jo Guldi, “The Common Landscape of Digital History,” Digital Histories: Emergent Approaches within the New Digital History (Helsinki: Helsinki University Press, 2020); Ashley S. Lee, Poom Chiarawongse, Jo Guldi, and Andras Zsom, “The Role of Critical Thinking in Humanities Infrastructure: The Pipeline Concept with a Study of HaToRI (Hansard Topic Relevance identifier),” Digital Humanities Quarterly 014:3 (2020). Jo Guldi, “The Measures of Modernity.

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16. Dzovinar Kévonian和Philippe Rygiel，" 连接的Ogres:数字时代的全球来源，" 世界21:1(s)2022年5月11日:73-79;Jo Guldi，" 数字历史的共同景观，" 数字历史:新数字历史中的紧急方法(赫尔辛基:赫尔辛基大学出版社，2020年);Ashley S. Lee，Poom Chiarawongse，Jo Guldi，和Andras Zsom，" 批判性思考在人文基础设施中的作用:带有哈托里研究的管道概念(汉萨德主题相关性标识)，" 数字人文季刊014:3(2020年)。 Jo Guldi，“现代化的尺度。

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Word Counts, Text Mining and the Promise and Limits of Present Tools as Indices of Historical Change,” International Journal for History, Culture and Modernity 7 (November 3, 2019), https://doi.org/10.18352/hcm.589; Jo Guldi, “The Modern Paradigms of Explanation: Significance, Agency, and Writing History in the Era of Climate Change,” Isis 110:2 (2019): 346-53; Ashley Lee, Jo Guldi, and Andras Zsom, “Measuring Similarity: Computationally Reproducing the Scholar’s Interests,” ArXiv:1812.05984 [Cs], December 14, 2018, http://arxiv.org/abs/1812.05984; Jo Guldi and Benjamin Williams, “Synthesis and Large-Scale Textual Corpora: A Nested Topic Model of Britain’s Debates over Landed Property in the Nineteenth Century,” Current Research in Digital History 1 (2018), https://doi.org/10.31835/crdh.2018.01; For validation, Jo Guldi, “Critical Search: A Procedure for Guided Reading in Large-Scale Textual Corpora,” Journal of Cultural Analytics (2018): 1-35, https://doi.org/10.22148/16.030.

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ENGLISH ORIGINAL

17. The wider field of the digital humanities includes practices such as metadata creation, mapping, network analysis, and the creation of online experiences and games. Cultural analytics focuses on the analysis of data - typically textual, musical, or visual - to produce insights about humanities questions, for instance the nature of cultural bias or the reality of change over time. For the digital humanities more broadly, see Eileen Gardiner and Ronald G. Musto, The Digital Humanities: A Primer for Students and Scholars (Cambridge University Press, 2015); for cultural analytics, see Lev Manovich, “The Science of Culture? Social Computing, Digital Humanities and Cultural Analytics,” CA: Journal of Cultural Analytics, May 23, 2016, http://culturalanalytics.org/2016/05/the-science-of-culture-social-computing-digital-humanities-and-cultural-analytics.

中文译文

17. 数字人文的更广泛领域包括元数据的生成、绘图、网络分析以及在线经验和游戏的创建。 文化分析的重点是分析数据，通常是文字、音乐或视觉数据，以产生对人文问题的见解，例如文化偏见的性质或随时间推移而变化的现实。 关于更广泛的数字人文，见Eileen Gardiner和Ronald G. Musto，《数字人文:学生和学者的入门》(剑桥大学出版社，2015年);关于文化分析，见Lev Manovich，“文化科学? 社会计算，数字人文与文化分析，"CA:文化分析杂志，2016年5月23日，http://culturalanalytics.org/2016/05/the-science-of-culture-social-computing-digital-humanities-and-cultural-analytics.

ENGLISH ORIGINAL

For a general introduction to the digital humanities, see Johanna Drucker, The Digital Humanities Coursebook (London: Routledge, 2021), and the various editions of Lauren Klein and Matthew Gold, Debates in the Digital Humanities (Minnesota, MN: University of Minnesota Press, 2004-20). For the long tradition of quantitative measures being used for literary criticism in the past, see Brad Pasanek, “Extreme Reading: Josephine Miles and the Scale of the Pre-Digital Digital Humanities,” ELH 86: 2 (2019): 369, 375. The digital social sciences, by contrast, have tended to come from departments of sociology and political science and they are experiencing their own revolution in text mining applied to traditional questions motivated by social theory. See Peter Bearman, “Big Data and Historical Social Science,” Big Data & Society 2:2 (December 27, 2015), https://doi.org/10.1177/2053951715612497; James A. Evans and Pedro Aceves, “Machine Translation: Mining Text for Social Theory,” Annual Review of Sociology 42:1 (2016): 21-50.

中文译文

关于数字人文的一般介绍，见Johanna Drucker,The Digital Humanities Course Book(伦敦:Routledge,2021),以及Lauren Klein和Matthew Gold的各种版本，数字人文的辩论(Minnesota,MN:明尼苏达大学出版社，2004-20). 关于过去用于文学批评的量化度量衡的悠久传统，见布拉德· 帕萨内克，"极限阅读:约瑟芬· 迈尔斯与数字前人文学科的规模，"ELH 86:2 (2019):369,375. 相比之下，数字社会科学往往来自社会学和政治学系，它们正在经历自己的革命，即运用于社会理论所激发的传统问题的文本挖掘。 见Peter Bearman,“大数据和历史社会科学”,“大数据和社会2:2(2015年12月27日) ,https://doi.org/10.1177/2053951715612497; James A. Evans和Pedro Aceves,“Machine 翻译:采矿文字用于社会理论”,《社会学年度评论》42:1(2016年):21-50。

ENGLISH ORIGINAL

The discipline of history has traditionally claimed membership both in the humanities and the social sciences, and typically draws on the methods of both parts of the university. For a general introduction to the variety of approaches used in digital history, including maps, images, and text from the past, see Hannu Salmi, What Is Digital History? (London: Polity, 2020). In this book I will invoke the word “humanities” broadly for interpretive issues in contrast to quantitative ones, even though many of the approaches I foreground are of interest to the social sciences or were developed within social science frameworks.

中文译文

历史学科传统上主张在人文科学和社会科学两方面都有会员资格，并一般借鉴大学两部分的方法。关于数字历史中采用的各种方法的一般介绍，包括地图、图像和过去的文本，见Hannu Salmi,《数字历史是什么? (伦敦: Polity, 2020). 在这本书中，我将泛泛地引用“人性”一词来解释与数量问题相对应的问题，尽管我展望的许多方法对社会科学感兴趣，或是在社会科学框架内制定的。

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18. As science and technology scholars will note, “hybridization” is, like “mining,” a deliberately ambivalent metaphor. Hybrid grains and the industrialized farms they supported drove smaller farmers from business in many parts of the world, helping to produce a global housing crisis and making agriculture overall more vulnerable - even if it was also more productive at scale. A monocrop of hybrid digital history would result in similar imbalances in the name of scale. There is a great deal more to be said that will not be said here. My intention in this volume is to show that hybrid practice can be done in a more responsible way, but I also underscore the need for the perseverance of traditional history and the teaching and practice of traditional methods.

中文译文

18. 正如科学和技术学者将指出的，" 混杂化 " 与 " 采矿 " 一样，是一种故意矛盾的比喻。 他们支持的混合谷物和工业化农场将世界许多地方的小农赶出了商业，帮助造成了全球住房危机，使农业整体上更加脆弱，即使农业规模更大。 混合数字历史的单曲会导致规模名称的类似不平衡。还有很多话不能在这里说。 我在本卷中的意图是表明，混合做法可以更加负责地进行，但我还强调，必须坚持传统历史和传统方法的教学和实践。

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19. Ted Underwood reviews this characterization, which he credits to Drew Conway, CEO of Alluvium, in a blog entry: https://tedunderwood.com/2017/01/29/two-syllabi-digital-humanitiesand-data-science-in-the-humanities.

中文译文

19. Ted Underwood在博客上评论了这个描述，https://tedunderwood.com/2017/01/29/two-syllabi-digital-humanitiesand-data-science-in-the-humanities.

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20. Tarleton Gillespie, The Relevance of Algorithms (Cambridge, MA: MIT Press, 2014), https://academic.oup.com/mit-press-scholarship-online/book/14976/chapter/169333383; Cathy O'Neil, Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy (New York: Crown, 2016); Safiya Umoja Noble, Algorithms of Oppression: How Search Engines Reinforce Racism (New York: NYU Press, 2018); Wendy Hui Kyong Chun, Discriminating Data: Correlation, Neighborhoods, and the New Politics of Recognition (Cambridge, MA: MIT Press, 2021); Kate Crawford, The Atlas of AI.

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20. Tarleton Gillespie，《算术的相关性》(坎布里奇，MA:麻省理工学院出版社，2014年)，https://academic.oup.com/mit-press-scholarship-online/book/14976/chapter/169333383; Cathy O ' Neil，《数学破坏武器:大数据如何增加不平等和威胁民主》(纽约:王室，2016年);Safiya " 团结 " 项目Noble，《压迫的算术:搜索引擎如何加强种族主义》(纽约:NYU出版社，2018年);Wendy Hui Kyong Chun，《辩驳数据:关联、邻居和承认的新政治》(坎布里奇，MA:麻省理工学院出版社，2021年);Kate Craword，《大赦国际的地图集》。

ENGLISH ORIGINAL

21. For my investigations of bias in the historical performance of GPT-3, see "The Dangerous Art of Text Mining," talk at the Alan Turing Institute, London, UK (July 2022), www.youtube.com/watch?v=-V7WZaUk5oA. I've applied it to summarizing my own books, and found that there are errors. Where I set out formal arguments, reviewing the historiography of the field and then stating my own position as a refutation of those arguments, GPT- 3 is just as likely to summarize the texts that I'm refuting as fact. It has no preference for my own argument or the evidence I've presented; it will summarize anything. It can summarize them with alacrity and questionable accuracy.

中文译文

21. 关于我对GPT-3历史表现的偏见的调查，见联合王国伦敦Alan Turing研究所(2022年7月)的 " 文字采矿的危险艺术 " 谈话，我用www.youtube.com/watch?v=-V7WZaUk5oA.来总结我自己的书，发现有错误。 GPT - 3同样有可能总结我所反驳的文本。它不赞成我自己的论点或我提出的证据，它可以轻松和令人怀疑的准确性来总结它们。

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22. Megan Morreale, "Daily SMS Mobile Usage Statistics," SMSEagle, March 6, 2017, www.smseagle.eu/2017/03/06/daily-sms-mobile-statistics.

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22. Megan Morreale, "每日短信移动使用统计，" SMSEagle, 2017年3月6日，www.smseagle.eu/2017/03/06/daily-sms-mobile-statistics.

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23. For literacy, see UNESCO Institute for Statistics, uis.unesco.org. For emails, see J. Clement, "Daily Number of E-Mails Worldwide 2023," Statista, August 9, 2019, www.statista.com/statistics/456500/daily-number-of-e-mails-worldwide.

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23. 关于识字率，见教科文组织统计研究所，uis.unesco.org。 关于电子邮件，见J. Clement，"2023年世界电子邮件日数"，Statista,2019年8月9日，www.statista.com/statistics/456500/daily-number-of-e-mails-worldwide.

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24. George Harwood and Evangeline Walker, "How Much of the Amazon Would It Take to Print the Internet?," Journal of Interdisciplinary Science 4 (2015): 10-11.

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24. George Harwood和Evangeline Walker，" 亚马逊岛要花多少钱印刷互联网？"，《跨学科科学杂志》4 (2015年):10-11。

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25. Executive Office of the President, Social and Behavioral Sciences Team, Annual Report(2015): 46 (with thanks to J. Nathan Matias), https://sbst.gov/download/2015%20SBST%20Annual%20Report.pdf?fbclid=IwAR1_A_UvH6V5zPzCP1FxJzJB4zsKwp8H7bNcoO3qVIgta5V-EgZsIUtRI9c.

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25. 总统办公厅，社会和行为科学小组，年度报告(2015):46(感谢J.Nathan Matias)，https://sbst.gov/download/2015%20SBST%20Annual%20Report.pdf? fbclid=IwAR1 A UvH6V5zPzCP1FxJzJB4zsKwp8H7bNcoO3qVIgta5V-EgZsIUtRI9c。

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26. For 18 billion pages, see http://opm.gov/policy-data-oversight/data-analysis-documentation/federal-employment-reports/historical-tables/total-government-employment-since-1962. For quotation, see Erez Aiden and Jean-Baptiste Michel, Uncharted: Big Data as a Lens on Human Culture (New York: Riverhead Books, 2013): 11. Aiden and Michel refer to big data in general rather than the problem of text.

中文译文

26. 180亿页，见http://opm.gov/policy-data-oversight/data-analysis-documentation/federal-employment-reports/historical-tables/total-government-employment-since-1962. 引文参见艾雷兹· 艾登和让-巴蒂斯特· 米歇尔，"未知:大数据作为关于人类文化的连载"(纽约:河头书社，2013年):11. 艾登和米歇尔泛指大数据，而不是文字问题.